

SOFTWARE REQUIREMENTS SPECIFICATION

Industry Intelligence Bridge (IIB)

AI-Powered Job & Internship Intelligence Platform

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Document Revision History

Version	Date	Author	Description of Change
0.1	2026-06-20	IIB Founding Team	Initial architecture and role-hierarchy discussion; monorepo and unified-model decisions.
0.2	2026-07-01	IIB Founding Team	College Module Feature Design Document (19 features, 20-section spec) completed.
0.3	2026-07-10	IIB Founding Team	Company Admin and Recruiter feature specifications incorporated.
1.0	2026-07-15	IIB Founding Team	First consolidated Software Requirements Specification issued, reconciling FDD tech stack (React/Express/MySQL/Gemini) as the system of record.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Industry Intelligence Bridge (IIB), a centralized job and internship intelligence platform connecting Colleges, Companies, and independent External Students. This document is intended to serve as the authoritative reference for engineering, QA, and stakeholder sign-off during the build of the IIB platform, and supersedes informal specifications where a conflict exists.

1.2 Document Conventions

Requirement statements use the keywords SHALL (mandatory), SHOULD (recommended, not mandatory), and MAY (optional), consistent with IEEE 830 convention. Functional requirements are identified with the prefix FR-, non-functional requirements with NFR-, business rules with BR-, and use cases with UC-, each using a sequential three-digit identifier. Every requirement in this document is written to be independently testable.

1.3 Intended Audience

- Engineering team (frontend, backend, database, DevOps) implementing the platform.
- Product and founding team (Gaurav Bhosale, Amey Nadkar, Dheeraj Sarfare, Aditee Talekar) validating scope and priority.
- Quality Assurance, for derivation of test cases and acceptance criteria.
- Future technical hires and external auditors/investors requiring a system-of-record specification.

1.4 Project Scope

IIB aggregates job and internship opportunities, verifies and categorizes them, and exposes them through role-based portals to seven user classes organized under College, Company, and platform hierarchies (see Section 2.3). The platform provides AI-augmented resume intelligence and recommendation features that are strictly additive to core workflows: an AI provider outage shall never block a user's ability to search, apply, review, or moderate. Excel and Google Sheets integrations are supported as derived, synchronizable exports and imports, never as the system of record. Third-party job-board scraping is explicitly out of scope for the initial release (MVP) and is deferred to Phase 2 (Section 12.3); direct company- and recruiter-submitted postings are the MVP's primary listing-acquisition path.

1.5 References

- IIB College Module Feature Design Document (FDD) — Module 1: College Admin, Placement Coordinator, College Student (19 features, 20-section production specification).
- IIB Platform Risk Management Report.
- IIB India competitor feature analysis (FlexJobs, We Work Remotely, Wellfound, Remote OK, Himalayas, and other remote/job-board platforms).
- IIB India ADAG Weekly Delivery Timeline.
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications (structural convention reference).

2. Overall Description

2.1 Product Perspective

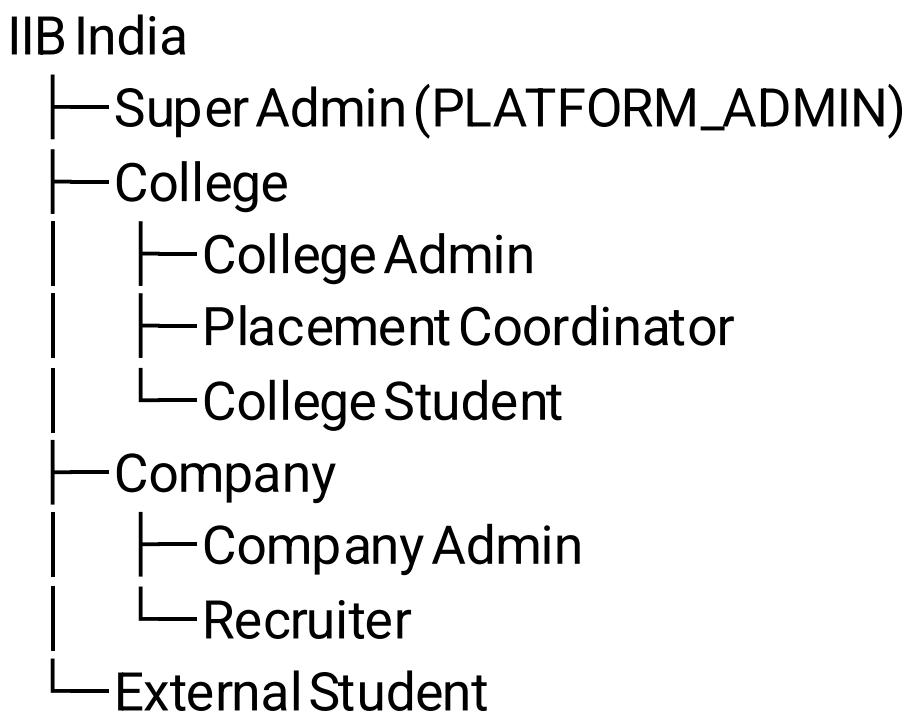
IIB is a new, standalone SaaS platform; it is not an extension of an existing internal system. It occupies a distinct position relative to two categories of existing products: (a) generalist and remote-work job boards (e.g., We Work Remotely, Wellfound, Remote OK, Himalayas) that serve individual job-seekers without a campus-placement layer, and (b) college placement-cell tools that typically operate as spreadsheet-driven, manual processes without applicant-facing discovery or AI augmentation. IIB's differentiated position is the integration of a campus placement workflow (College Admin → Placement Coordinator → Student) with a standard employer/recruiter hiring workflow (Company Admin → Recruiter) inside one platform, plus an AI services layer common to both.

2.2 Product Functions

- Role-based account provisioning and access control across seven user classes.
- Listing submission, moderation, and publication for jobs and internships.
- Opportunity discovery via search, filtering, and AI-assisted recommendation.
- Application submission and end-to-end tracking through to hiring outcome.
- Resume upload, parsing, scoring, and AI-assisted improvement, with graceful non-AI fallback.
- Placement and hiring analytics and reporting, including Excel/CSV export and Excel/Google Sheets synchronization.
- Platform-level moderation, auditing, and AI-service configuration for the Super Admin.

2.3 User Classes

The platform defines seven user classes organized into three hierarchies, illustrated below.



The Super Admin (internally named PLATFORM_ADMIN in code and schema to avoid confusion with Placement Coordinator) represents the platform's own developer/operations team and is not a placement-adjacent function; it has cross-tenant visibility that no other role holds.

2.4 Operating Environment

- Client: modern evergreen web browsers (Chrome, Edge, Firefox, Safari), responsive down to 360px viewport width.
- Frontend: React.js single-page/server-rendered application.
- Backend: Node.js with Express.js, exposing a REST API.
- Database: MySQL, accessed via the Prisma ORM.

- Cache/Queue: Redis, used for caching and rate-limit state.
- AI Services: Google Gemini, isolated behind an internal AI services module.
- Deployment: Docker containers on a compatible host, or a Vercel (frontend) + Render (backend) + managed/Railway MySQL topology.

2.5 Constraints

- Third-party scraping (Puppeteer/Cheerio/Playwright) is technically available but is deferred past MVP by product decision; it must not be relied upon for MVP listing volume.
- AI features depend on an external provider (Google Gemini) and must degrade gracefully rather than block core workflows on provider outage or latency.
- Excel and Google Sheets are supported as derived exports/synchronized artifacts only; they must never become an upstream source of truth for platform data.
- Multi-tenant data isolation (college_id / company_id scoping) must be enforced on every query; this is a hard security constraint, not a UI-layer convenience.

2.6 Assumptions

- Colleges and companies onboarding to the platform have institutional email domains suitable for identity verification.
- Initial listing volume will be sufficient without third-party scraping, given direct recruiter/company submission and college partnerships.
- Google Gemini's API availability and terms of service remain suitable for production use throughout the initial release cycle.

3. System Features and Functional Requirements

This chapter enumerates functional requirements grouped by role, sub-role, and functional module, consistent with the platform's role hierarchy (Section 2.3). Each requirement carries a unique identifier, a priority (Shall/Should), and a testable SHALL-form description. Where a module has an approved Feature Design Document (FDD) with full production-level specification (frontend, backend, database, API, and security design), this is noted; the FDD is authoritative for implementation detail, and this SRS is authoritative for requirement scope and traceability.

3.1 College Role

The College Role encompasses three sub-roles operating within a single tenant (a college/institution): College Admin, Placement Coordinator, and College Student. All College Role data is isolated per institution via tenant scoping (college_id) enforced at both the application and database layers.

3.1.1 College Admin

The College Admin is the top-level administrator for a college tenant. The College Admin provisions Placement Coordinator and Student accounts, manages departments, configures the college's profile and platform settings, controls role-based permissions, and consumes placement reporting. This sub-role corresponds to Module 1, Features 1-7 of the IIB College Module Feature Design Document (FDD).

3.1.1.1 Placement Coordinator Account Management

Enables the College Admin to create, edit, suspend, and remove Placement Coordinator accounts and to assign each coordinator to one or more departments via a token-based email invite workflow.

Req ID	Priority	Description
FR-001	Shall	The system shall allow a College Admin to create a Placement Coordinator account with name, email, and one or more department assignments.
FR-002	Shall	The system shall send a single-use, time-limited (72-hour) email invite token to a newly created Placement Coordinator to activate their account.
FR-003	Should	The system shall allow a College Admin to edit a Placement Coordinator's name and department assignments.
FR-004	Shall	The system shall allow a College Admin to suspend or reactivate a Placement Coordinator account.
FR-005	Should	The system shall allow a College Admin to remove (soft-delete) a Placement Coordinator account, blocking removal if the coordinator has pending moderation actions assigned.
FR-006	Should	The system shall allow a College Admin to resend an invite, invalidating any prior unused invite token.
FR-007	Shall	The system shall reject duplicate Placement Coordinator email addresses within the same college tenant.

3.1.1.2 Student Account Management (Individual & Bulk)

Enables the College Admin to onboard College Students individually or via bulk CSV/Excel import, and to manage their department affiliation and account status.

Req ID	Priority	Description
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Req ID	Priority	Description
FR-008	Shall	The system shall allow a College Admin to create an individual College Student account with name, email, enrollment ID, and department.
FR-009	Shall	The system shall allow a College Admin to bulk-import College Student accounts via a CSV/Excel upload, validating each row before commit.
FR-010	Shall	The system shall report row-level validation errors from a bulk import without discarding valid rows.
FR-011	Shall	The system shall allow a College Admin to edit a student's department, batch/year, and contact details.
FR-012	Shall	The system shall allow a College Admin to block or remove a student account.
FR-013	Shall	The system shall allow colleges to mark a student as verified/enrolled to distinguish genuine students from unverified self-registrations.

3.1.1.3 Department Management

Enables the College Admin to define and maintain the department taxonomy used for student, coordinator, and reporting scoping.

Req ID	Priority	Description
FR-014	Shall	The system shall allow a College Admin to create, rename, and archive departments.
FR-015	Shall	The system shall prevent hard deletion of a department that is referenced by any student, coordinator, or listing record; departments in use shall only be archivable.
FR-016	Shall	The system shall allow a College Admin to view a count of students and coordinators per department.

3.1.1.4 College Profile, Branding & Platform Settings

Enables the College Admin to maintain the institution's public profile, branding assets, and college-level platform configuration.

Req ID	Priority	Description
FR-017	Shall	The system shall allow a College Admin to edit the college profile, including name, address, accreditation details, and website.
FR-018	Shall	The system shall allow a College Admin to upload a college logo and set brand colors used on college-facing pages.
FR-019	Shall	The system shall allow a College Admin to configure college-level platform settings, including default application deadlines and listing visibility windows.

3.1.1.5 Roles, Permissions & Access Control

Enables the College Admin to define fine-grained permissions for Placement Coordinators and to review effective access at any time.

Req ID	Priority	Description
FR-020	Shall	The system shall allow a College Admin to assign granular permissions (e.g., approve-listings, export-data, manage-students) to a Placement Coordinator independent of department scope.

Req ID	Priority	Description
FR-021	Shall	The system shall enforce that every College Admin and Placement Coordinator action is authorized against the caller's role and permission set on every request, not only at the UI layer.
FR-022	Shall	The system shall allow a College Admin to view an access-control summary listing each coordinator's effective permissions.

3.1.1.6 Reports & Analytics Export

Enables the College Admin to export placement reporting in Excel and CSV formats, scoped by department and reporting period.

Req ID	Priority	Description
FR-023	Shall	The system shall allow a College Admin to export a department-wise placement report in Excel (.xlsx) format.
FR-024	Shall	The system shall allow a College Admin to export placement data in CSV format.
FR-025	Shall	The system shall allow a College Admin to generate a monthly placement summary report.
FR-026	Shall	The system shall allow a College Admin to generate an annual placement summary report.
FR-027	Shall	The system shall generate all exports from live database records, treating Excel/CSV output strictly as a derived artifact, never as a system of record.

3.1.1.7 Notification Settings & Engagement Triggers

Enables the College Admin to configure which platform-wide notification events are active for their college tenant.

Req ID	Priority	Description
FR-028	Shall	The system shall allow a College Admin to enable or disable college-level notification categories (e.g., student-selected, internship-started).
FR-029	Shall	The system shall notify a College Admin when a student under their college is marked as Selected by a company.
FR-030	Shall	The system shall notify a College Admin when a student under their college begins an internship.
FR-031	Should	The system shall allow a College Admin to configure the notification channels (email, in-app) used for each category.

3.1.2 Placement Coordinator

The Placement Coordinator is the operational workforce role responsible for reviewing and moderating job/internship listings, tracking student applications and outcomes, generating recommendation letters, and producing placement analytics. This sub-role corresponds to Module 1, Features 8-13 of the FDD.

3.1.2.1 Listing Review & Moderation Workflow

Enables the Placement Coordinator to review, approve, reject, edit, archive, and pin job and internship listings surfaced to their college's students.

Req ID	Priority	Description
FR-032	Shall	The system shall present newly submitted job listings to Placement Coordinators for review before publication to students.
FR-033	Shall	The system shall present newly submitted internship listings to Placement Coordinators for review before publication to students.
FR-034	Shall	The system shall allow a Placement Coordinator to approve a listing, making it visible to eligible students.
FR-035	Shall	The system shall allow a Placement Coordinator to reject a listing with a mandatory reason, notifying the submitting Recruiter/Company Admin.
FR-036	Shall	The system shall allow a Placement Coordinator to edit listing metadata (deadline, eligibility criteria) prior to approval.
FR-037	Shall	The system shall allow a Placement Coordinator to archive an expired or withdrawn listing.
FR-038	Should	The system shall allow a Placement Coordinator to pin a listing to the top of the student-facing feed.

3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore

Enables the Placement Coordinator to export and synchronize placement data with Excel and Google Sheets, and to back up and restore data snapshots. Excel and Google Sheets integrations are strictly one-directional or reconciled derived exports; the platform database remains the single source of truth.

Req ID	Priority	Description
FR-039	Shall	The system shall allow a Placement Coordinator to export current placement data to Excel on demand.
FR-040	Shall	The system shall allow a Placement Coordinator to export current placement data to Google Sheets on demand.
FR-041	Shall	The system shall support a scheduled auto-sync of placement data to a connected Excel file at a coordinator-configured interval.
FR-042	Shall	The system shall support a scheduled auto-sync of placement data to a connected Google Sheet at a coordinator-configured interval.
FR-043	Shall	The system shall allow a Placement Coordinator to create a manual backup snapshot of college placement data in Excel format.
FR-044	Shall	The system shall allow a Placement Coordinator to restore college placement data from a prior backup snapshot, with a confirmation step and audit log entry.
FR-045	Shall	The system shall never treat an imported Excel or Google Sheets file as authoritative; any conflicting inbound edits shall require explicit coordinator confirmation before being applied to the database.

3.1.2.3 Letter of Recommendation (LOR) Generator

Enables the Placement Coordinator to generate, customize, and export Letters of Recommendation for students.

Req ID	Priority	Description
FR-046	Shall	The system shall allow a Placement Coordinator to generate a Letter of Recommendation for a student from a standard template pre-filled with student and placement data.

Req ID	Priority	Description
FR-047	Shall	The system shall allow a Placement Coordinator to edit the generated LOR content before finalizing it.
FR-048	Shall	The system shall allow a Placement Coordinator to export a finalized LOR as a signed PDF document.
FR-049	Should	The system shall maintain a history of LORs issued per student for audit purposes.

3.1.2.4 Student & Application Tracking Console

Enables the Placement Coordinator to view students, track their applications, and monitor placement and internship status department-wise.

Req ID	Priority	Description
FR-050	Shall	The system shall allow a Placement Coordinator to view the list of students within their assigned department(s).
FR-051	Shall	The system shall allow a Placement Coordinator to track the application status of any student under their scope, from applied through to selected/rejected.
FR-052	Shall	The system shall allow a Placement Coordinator to view a student's overall placement status.
FR-053	Shall	The system shall allow a Placement Coordinator to view a student's internship status.
FR-054	Shall	The system shall allow a Placement Coordinator to view a student's uploaded resume, subject to resume-access logging.
FR-055	Shall	The system shall allow a Placement Coordinator to filter the student list by department.

3.1.2.5 Coordinator Notification Hub

Delivers event-driven notifications to the Placement Coordinator across the student application and placement lifecycle.

Req ID	Priority	Description
FR-056	Shall	The system shall notify a Placement Coordinator when a student under their scope applies to a listing.
FR-057	Shall	The system shall notify a Placement Coordinator when a student is shortlisted.
FR-058	Shall	The system shall notify a Placement Coordinator when a student is selected.
FR-059	Shall	The system shall notify a Placement Coordinator when a student's internship starts.
FR-060	Shall	The system shall notify a Placement Coordinator when a student's internship is completed.
FR-061	Shall	The system shall notify a Placement Coordinator when a student receives a Pre-Placement Offer (PPO).
FR-062	Shall	The system shall notify a Placement Coordinator when a student accepts a job offer.

3.1.2.6 Placement Analytics Dashboard & Unified Search

Provides the Placement Coordinator with placement KPIs and a unified search across students, companies, jobs, and internships.

Req ID	Priority	Description
FR-063	Shall	The system shall display the count of students placed, scoped by department and time period.
FR-064	Shall	The system shall display company hiring trends for the coordinator's college.
FR-065	Shall	The system shall display department-wise placement performance.
FR-066	Shall	The system shall display the highest package offered within a selected period.
FR-067	Shall	The system shall display the average package offered within a selected period.
FR-068	Shall	The system shall display the count of currently active opportunities visible to the college.
FR-069	Shall	The system shall allow a Placement Coordinator to search for a student by name, enrollment ID, or department.
FR-070	Shall	The system shall allow a Placement Coordinator to search for a company by name.
FR-071	Shall	The system shall allow a Placement Coordinator to search for an internship listing.
FR-072	Shall	The system shall allow a Placement Coordinator to search for a job listing.

3.1.3 College Student

The College Student is a verified student affiliated with a college tenant, discovering and applying to opportunities, managing a resume, and maintaining a career profile. This sub-role corresponds to Module 1, Features 14-19 of the FDD.

3.1.3.1 Personalized Dashboard & AI Recommendations

Provides the College Student with a personalized landing view surfacing recommended and time-sensitive opportunities.

Req ID	Priority	Description
FR-073	Shall	The system shall display a personalized dashboard summarizing recommended opportunities, saved items, and application status for the logged-in student.
FR-074	Shall	The system shall display AI-recommended job listings based on the student's profile and resume, with a rule-based fallback ranking if the AI service is unavailable.
FR-075	Shall	The system shall display AI-recommended internship listings based on the student's profile and resume, with a rule-based fallback ranking if the AI service is unavailable.
FR-076	Shall	The system shall display recently added listings relevant to the student's department and eligibility.
FR-077	Shall	The system shall display listings closing within a configurable threshold (e.g., 72 hours).

3.1.3.2 Job/Internship/Company Search & Filters

Enables the College Student to search and filter jobs, internships, and companies.

Req ID	Priority	Description
FR-078	Shall	The system shall allow a College Student to search job listings by keyword.
FR-079	Shall	The system shall allow a College Student to search internship listings by keyword.
FR-080	Shall	The system shall allow a College Student to search companies by name.

Req ID	Priority	Description
FR-081	Shall	The system shall allow a College Student to filter listings by domain/field.
FR-082	Shall	The system shall allow a College Student to filter listings by location.
FR-083	Shall	The system shall allow a College Student to filter job listings by salary range.
FR-084	Shall	The system shall allow a College Student to filter internship listings by stipend range.
FR-085	Shall	The system shall allow a College Student to filter listings by required experience level.
FR-086	Shall	The system shall allow a College Student to filter listings by work mode (remote/hybrid/on-site).
FR-087	Shall	The system shall allow a College Student to filter listings by company.

3.1.3.3 Resume Management Suite

Enables the College Student to upload, analyze, build, and version their resume.

Req ID	Priority	Description
FR-088	Shall	The system shall allow a College Student to upload a resume in PDF or DOCX format.
FR-089	Shall	The system shall provide an AI-generated resume analysis summarizing strengths and gaps, degrading gracefully to a checklist-based analysis if the AI service is unavailable.
FR-090	Shall	The system shall compute and display a resume score based on completeness and formatting heuristics.
FR-091	Should	The system shall provide a set of resume templates the student may apply to a builder-generated resume.
FR-092	Shall	The system shall provide a guided resume builder that produces a downloadable resume document.
FR-093	Shall	The system shall retain prior resume versions and allow a student to restore or compare versions.

3.1.3.4 Apply / Save / Bookmark & Application Tracking

Enables the College Student to apply to opportunities and track application status over time.

Req ID	Priority	Description
FR-094	Shall	The system shall allow a College Student to apply to an approved, eligible job or internship listing.
FR-095	Shall	The system shall allow a College Student to save a listing for later review without applying.
FR-096	Shall	The system shall allow a College Student to bookmark a listing.
FR-097	Shall	The system shall allow a College Student to track the status of each submitted application.
FR-098	Shall	The system shall display a chronological application history for the student.

3.1.3.5 Student Profile

Enables the College Student to maintain a structured career profile.

Req ID	Priority	Description
FR-099	Shall	The system shall allow a College Student to list and edit skills on their profile.
FR-100	Shall	The system shall allow a College Student to enter education details, pre-populated from college enrollment data where available.
FR-101	Shall	The system shall allow a College Student to list projects with descriptions and links.
FR-102	Shall	The system shall allow a College Student to list certificates with issuing authority and date.
FR-103	Should	The system shall allow a College Student to list career interests used to inform recommendations.

3.1.3.6 Company Reviews, Salary Insights & Interview Experience

Provides the College Student with company research signals aggregated from platform and third-party sources.

Req ID	Priority	Description
FR-104	Shall	The system shall display aggregated company reviews submitted by verified platform users.
FR-105	Shall	The system shall display salary insights for a company or role where sufficient data exists.
FR-106	Shall	The system shall allow a College Student to view and submit interview experience write-ups tied to a specific company and role.

3.2 Company Role

The Company Role encompasses two sub-roles operating within a single company tenant: Company Admin and Recruiter. Company Role data is isolated per organization via tenant scoping (company_id).

3.2.1 Company Admin

The Company Admin manages the company's organizational presence on the platform, oversees job/internship listings, reviews hiring analytics, and manages Recruiter accounts.

3.2.1.1 Company Management

Enables the Company Admin to maintain the company's verified organizational profile.

Req ID	Priority	Description
FR-107	Shall	The system shall allow a Company Admin to create and edit the company profile, including industry, size, and description.
FR-108	Shall	The system shall allow a Company Admin to manage one or more branch/office locations under the company profile.
FR-109	Shall	The system shall require a Company Admin to submit GST/CIN and an official business email during onboarding, with the account held in a pending state until Super Admin approval.
FR-110	Shall	The system shall allow a Company Admin to upload company branding assets (logo, cover image) for the public company profile.

3.2.1.2 Job & Internship Listing Management

Enables the Company Admin to manage listings at an organizational level, independent of the individual Recruiter who created them.

Req ID	Priority	Description
FR-111	Shall	The system shall allow a Company Admin to create job listings.
FR-112	Shall	The system shall allow a Company Admin to create internship listings.
FR-113	Shall	The system shall allow a Company Admin to edit any listing owned by the company.
FR-114	Shall	The system shall allow a Company Admin to delete a listing owned by the company.
FR-115	Shall	The system shall allow a Company Admin to archive an expired listing owned by the company.

3.2.1.3 Hiring Analytics & Reports

Provides the Company Admin with organization-wide visibility into listing performance and hiring outcomes.

Req ID	Priority	Description
FR-116	Shall	The system shall display view counts for each company listing.
FR-117	Shall	The system shall display the count of applicants per listing.
FR-118	Shall	The system shall display the count of shortlisted candidates per listing.
FR-119	Shall	The system shall display the count of selected candidates per listing.
FR-120	Shall	The system shall allow a Company Admin to export a consolidated hiring report.

3.2.1.4 Team & Recruiter Management

Enables the Company Admin to provision and manage Recruiter accounts under the company tenant.

Req ID	Priority	Description
FR-121	Shall	The system shall allow a Company Admin to add a Recruiter account, scoped to the company tenant.
FR-122	Shall	The system shall allow a Company Admin to remove a Recruiter account.
FR-123	Shall	The system shall allow a Company Admin to assign granular permissions to a Recruiter (e.g., post-listing, view-applicants only).

3.2.2 Recruiter

The Recruiter manages individual job and internship postings, screens and progresses candidates, schedules interviews, and communicates with applicants.

3.2.2.1 Job & Internship Posting Management

Enables the Recruiter to create and maintain individual listings.

Req ID	Priority	Description
FR-124	Shall	The system shall allow a Recruiter to post a new internship listing, subject to college/placement-coordinator review where the listing targets a specific college.
FR-125	Shall	The system shall allow a Recruiter to post a new job listing.

Req ID	Priority	Description
FR-126	Shall	The system shall allow a Recruiter to edit their own listing prior to its deadline.
FR-127	Shall	The system shall allow a Recruiter to close a listing before its scheduled deadline.
FR-128	Shall	The system shall allow a Recruiter to extend a listing's application deadline.

3.2.2.2 Candidate Management

Enables the Recruiter to review and progress applicants against a listing.

Req ID	Priority	Description
FR-129	Shall	The system shall allow a Recruiter to view all applicants to their listing.
FR-130	Shall	The system shall allow a Recruiter to filter applicants by criteria such as college, department, or resume score.
FR-131	Shall	The system shall allow a Recruiter to download an applicant's resume, and shall log every resume download with recruiter identity and timestamp.
FR-132	Shall	The system shall allow a Recruiter to shortlist a candidate.
FR-133	Shall	The system shall allow a Recruiter to reject a candidate.

3.2.2.3 Interview Management

Enables the Recruiter to schedule and record interview outcomes.

Req ID	Priority	Description
FR-134	Shall	The system shall allow a Recruiter to schedule an interview for a shortlisted candidate.
FR-135	Shall	The system shall allow a Recruiter to update the status of a scheduled interview.
FR-136	Shall	The system shall allow a Recruiter to mark a candidate as Selected.
FR-137	Shall	The system shall allow a Recruiter to mark a candidate as Rejected.
FR-138	Shall	The system shall allow a Recruiter to mark an offer as Released for a selected candidate.

3.2.2.4 Communication

Enables the Recruiter to message applicants individually or in bulk.

Req ID	Priority	Description
FR-139	Shall	The system shall allow a Recruiter to send an in-platform message to an applicant.
FR-140	Shall	The system shall allow a Recruiter to email a candidate directly from the platform.
FR-141	Shall	The system shall allow a Recruiter to send a bulk notification to all applicants of a given listing.

3.3 External Student Role

The External Student is an individual user not affiliated with any college tenant on the platform. External Students self-register, are not subject to college-level moderation, and have access to a broader set of

AI-augmented and premium features than College Students.

3.3.1 External Student

The External Student discovers opportunities, manages an AI-augmented resume and career profile, researches companies, and may subscribe to premium features for enhanced recommendations and analytics.

3.3.1.1 Dashboard & Discovery

Surfaces recommended and trending opportunities on the External Student's dashboard.

Req ID	Priority	Description
FR-142	Shall	The system shall display AI-recommended job listings on the External Student dashboard, with a rule-based fallback if the AI service is unavailable.
FR-143	Shall	The system shall display AI-recommended internship listings on the External Student dashboard, with a rule-based fallback if the AI service is unavailable.
FR-144	Shall	The system shall display trending job listings based on platform-wide application velocity.
FR-145	Shall	The system shall display trending internship listings based on platform-wide application velocity.
FR-146	Shall	The system shall display newly published opportunities matching the student's stated interests.

3.3.1.2 Search & Smart Filters

Provides comprehensive search and filtering across jobs, internships, and companies.

Req ID	Priority	Description
FR-147	Shall	The system shall allow an External Student to search job listings by keyword.
FR-148	Shall	The system shall allow an External Student to search internship listings by keyword.
FR-149	Shall	The system shall allow an External Student to search companies by name.
FR-150	Shall	The system shall allow an External Student to filter listings by required skill.
FR-151	Shall	The system shall allow an External Student to filter listings by domain.
FR-152	Shall	The system shall allow an External Student to filter job listings by salary range.
FR-153	Shall	The system shall allow an External Student to filter internship listings by stipend range.
FR-154	Shall	The system shall allow an External Student to filter listings restricted to fully remote work.
FR-155	Shall	The system shall allow an External Student to filter listings restricted to hybrid work.
FR-156	Shall	The system shall allow an External Student to filter listings by required experience level.
FR-157	Shall	The system shall allow an External Student to filter listings by city.
FR-158	Shall	The system shall allow an External Student to filter listings by company.

3.3.1.3 Resume Intelligence

Provides AI-augmented resume tooling for the External Student, all with non-AI fallback paths.

Req ID	Priority	Description
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Req ID	Priority	Description
FR-159	Shall	The system shall allow an External Student to upload a resume in PDF or DOCX format.
FR-160	Shall	The system shall parse an uploaded resume into structured fields (skills, education, experience).
FR-161	Shall	The system shall compute a resume score based on completeness and formatting heuristics.
FR-162	Shall	The system shall provide an AI-generated resume redesign suggestion, degrading gracefully to a static best-practices checklist if the AI service is unavailable.
FR-163	Shall	The system shall provide a guided resume builder that produces a downloadable resume document.
FR-164	Should	The system shall provide a set of resume templates for use in the resume builder.
FR-165	Shall	The system shall provide an ATS-compatibility check that flags formatting likely to fail automated parsing.

3.3.1.4 AI Career Assistant

Provides AI-driven career guidance; every capability in this module is additive and the platform's core application workflows shall remain functional if the AI service is degraded or unavailable.

Req ID	Priority	Description
FR-166	Shall	The system shall provide AI-generated job recommendations based on the student's profile and resume.
FR-167	Shall	The system shall provide AI-generated internship recommendations based on the student's profile and resume.
FR-168	Shall	The system shall provide an AI-generated skill-gap analysis comparing the student's profile against target roles.
FR-169	Should	The system shall provide AI-generated learning-resource suggestions to close identified skill gaps.
FR-170	Shall	The system shall provide an AI-generated career roadmap outlining suggested next steps.
FR-171	Shall	The system shall provide an AI-generated cover letter drafted for a specific listing, editable before submission.

3.3.1.5 Company Insights

Aggregates review, salary, and hiring-trend signals for company research.

Req ID	Priority	Description
FR-172	Shall	The system shall display aggregated company reviews submitted by verified platform users.
FR-173	Should	The system shall display third-party company review signals (e.g., Glassdoor) where a licensed data source is integrated; this is deferred past MVP pending third-party licensing.
FR-174	Should	The system shall display third-party company review signals (e.g., AmbitionBox) where a licensed data source is integrated; this is deferred past MVP pending third-party licensing.

Req ID	Priority	Description
FR-175	Should	The system shall display third-party company review signals (e.g., Indeed) where a licensed data source is integrated; this is deferred past MVP pending third-party licensing.
FR-176	Shall	The system shall display salary insights for a company or role where sufficient platform data exists.
FR-177	Shall	The system shall display hiring trend data for a company over a selected period.

3.3.1.6 Applications

Enables the External Student to apply to and track opportunities.

Req ID	Priority	Description
FR-178	Shall	The system shall allow an External Student to apply to an approved, eligible listing.
FR-179	Shall	The system shall allow an External Student to save a listing without applying.
FR-180	Shall	The system shall allow an External Student to bookmark a listing.
FR-181	Shall	The system shall allow an External Student to track application status.
FR-182	Shall	The system shall display an application timeline showing status changes with timestamps.

3.3.1.7 Premium Features

Gates an enhanced tier of AI and analytics capability behind a subscription, per the platform's Premium Feature Rules (Section 6.7). All premium features are additive; free-tier core functionality is never blocked by subscription status.

Req ID	Priority	Description
FR-183	Shall	The system shall provide enhanced AI resume optimization to subscribed premium students.
FR-184	Shall	The system shall provide an AI career coach conversational feature to subscribed premium students.
FR-185	Shall	The system shall provide unlimited job/internship alerts to subscribed premium students, versus a capped alert count for free-tier students.
FR-186	Shall	The system shall prioritize premium students' listings in AI recommendation ranking, clearly labeled as sponsored placement where applicable.
FR-187	Shall	The system shall provide advanced analytics (e.g., application-to-interview conversion rate) to subscribed premium students.
FR-188	Should	The system shall provide an extended resume version history to subscribed premium students beyond the free-tier retention limit.

3.4 Super Admin Role

The Super Admin (internally: PLATFORM_ADMIN) represents Anthropic-independent platform operations/developer staff and is explicitly distinct from any placement-adjacent function such as Placement Coordinator. The Super Admin has cross-tenant visibility and controls platform-wide configuration, moderation, data sourcing, and AI service management.

3.4.1 Super Admin (PLATFORM_ADMIN)

Operates at the platform level across all College and Company tenants.

3.4.1.1 Platform Management

Enables the Super Admin to manage tenant and user accounts across the platform.

Req ID	Priority	Description
FR-189	Shall	The system shall allow a Super Admin to view, approve, suspend, or remove College tenant accounts.
FR-190	Shall	The system shall allow a Super Admin to view, approve, suspend, or remove Company tenant accounts, including final approval of company GST/CIN verification.
FR-191	Shall	The system shall allow a Super Admin to view and, where necessary, suspend Student accounts across all tenants.
FR-192	Shall	The system shall allow a Super Admin to view and, where necessary, suspend Recruiter accounts across all tenants.
FR-193	Shall	The system shall allow a Super Admin to create and manage other platform-level admin accounts, with all such actions captured in the audit log.

3.4.1.2 Data Sources

Enables the Super Admin to configure listing acquisition sources. Per platform scope decisions, third-party scraping of external job boards is deferred past MVP; direct company/recruiter-submitted postings are prioritized first.

Req ID	Priority	Description
FR-194	Should	The system shall allow a Super Admin to configure and enable a scraper-based data source; this capability is deferred past MVP.
FR-195	Should	The system shall allow a Super Admin to configure a third-party API integration as a listing source; this capability is deferred past MVP.
FR-196	Should	The system shall allow a Super Admin to register a new external platform as a candidate data source; this capability is deferred past MVP.
FR-197	Shall	The system shall provide the Super Admin with monitoring of configured data source health and last-successful-sync status.

3.4.1.3 Moderation

Enables the Super Admin to review platform-wide content reports and maintain the listing taxonomy.

Req ID	Priority	Description
FR-198	Shall	The system shall allow a Super Admin to review listings and profiles flagged/reported by users.
FR-199	Shall	The system shall allow a Super Admin to remove a listing identified as spam or fraudulent.
FR-200	Shall	The system shall allow a Super Admin to manage the platform's listing category taxonomy.
FR-201	Shall	The system shall allow a Super Admin to manage the platform's domain/industry taxonomy.

3.4.1.4 Analytics

Provides the Super Admin with platform-wide growth and operational metrics.

Req ID	Priority	Description
FR-202	Shall	The system shall display platform-wide growth metrics over a selected period.
FR-203	Shall	The system shall display user growth metrics segmented by role.
FR-204	Shall	The system shall display the count of currently active job listings platform-wide.
FR-205	Shall	The system shall display the count of currently active internship listings platform-wide.
FR-206	Shall	The system shall display company growth metrics over a selected period.
FR-207	Shall	The system shall display college growth metrics over a selected period.
FR-208	Shall	The system shall display a revenue dashboard summarizing premium subscription and any other monetized activity.

3.4.1.5 Security

Provides the Super Admin with audit and monitoring capability across the platform.

Req ID	Priority	Description
FR-209	Shall	The system shall provide the Super Admin with a searchable audit log of all mutating actions platform-wide, including actor, action, target, and timestamp.
FR-210	Shall	The system shall provide the Super Admin with a login log capturing successful and failed authentication attempts.
FR-211	Shall	The system shall provide the Super Admin with an access log for sensitive resource reads (e.g., resume downloads).
FR-212	Shall	The system shall provide the Super Admin with API request monitoring, including error-rate and latency visibility.
FR-213	Shall	The system shall provide the Super Admin with backup management, including manual trigger and restore-point selection.

3.4.1.6 AI Management

Enables the Super Admin to configure and monitor the platform's isolated AI services module, keeping AI functionality decoupled from core platform availability.

Req ID	Priority	Description
FR-214	Shall	The system shall allow a Super Admin to configure the active AI model/provider used by the AI services module.
FR-215	Shall	The system shall allow a Super Admin to manage and version prompt templates used by AI features.
FR-216	Shall	The system shall allow a Super Admin to monitor recommendation-engine performance and override ranking weights.
FR-217	Shall	The system shall allow a Super Admin to monitor resume-engine (parsing/scoring) performance and accuracy.
FR-218	Shall	The system shall isolate all AI service calls in a dedicated module such that an AI

Req ID	Priority	Description
		provider outage does not block core, non-AI platform workflows.

Note on FDD alignment: Section 3.1 (College Role) requirements correspond directly to Features 1–19 of the IIB College Module FDD. Company Role, External Student Role, and Super Admin Role FDDs, at equivalent 20-section production depth, are recommended as the next documentation deliverable but are outside the scope of this SRS (see Section 12.1).

4. External Interface Requirements

4.1 User Interface

- The system shall provide a responsive web interface built in React.js, supporting viewport widths from 360px (mobile) through desktop.
- The system shall provide role-specific navigation and dashboards; a user shall never see navigation entries for actions their role/permissions do not grant.
- The system shall provide consistent form validation, loading, empty, and error states across all modules, per the pattern established in the College Module FDD (Section 4 of each feature).

4.2 Software Interface

- The system shall interface with the Google Gemini API for AI-augmented features (resume analysis, recommendations, cover-letter generation), isolated behind an internal AI services module.
- The system shall interface with an email delivery provider (e.g., SMTP/Resend) for transactional email (invites, notifications).
- The system shall interface with the Google Sheets API for Placement Coordinator data synchronization (Section 3.1.2, Feature: Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore).
- The system shall interface with an antivirus scanning service or library for uploaded file validation.

4.3 Hardware Interface

Not applicable. IIB is a cloud-hosted SaaS platform with no direct hardware interface requirements; client hardware is limited to standard consumer devices capable of running a modern web browser.

4.4 Communication Interface

- All client-server communication shall occur over HTTPS/TLS 1.2 or higher.
- The system shall expose a REST API over HTTP/JSON as the primary integration surface.
- Background job communication (e.g., BullMQ-equivalent queue processing, or an Express-compatible job-queue library) shall occur over the Redis protocol within the platform's private network.

4.5 API Interface

- The system shall expose versioned REST endpoints (e.g., /api/v1/...) for all platform functionality described in Chapter 3.
- Every API endpoint shall enforce JWT-based authentication and role/permission-based authorization, as specified in Section 5.2.
- The API shall return standard HTTP status codes (200/201/204 success; 400/401/403/404/409/422 client error; 500 server error) with a machine-readable error code and human-readable message on failure.

5. Non-Functional Requirements

5.1 Performance

Req ID	Priority	Description
NFR-001	Shall	The system shall return search and listing API responses within 500ms at the 95th percentile under nominal load (defined in Section 10.2).
NFR-002	Shall	The system shall complete an Excel/CSV export of up to 10,000 records within 15 seconds.
NFR-003	Shall	AI-augmented endpoints (recommendations, resume analysis) shall return within 5 seconds at the 95th percentile, or fall back to a non-AI default response within 2 seconds if the AI provider exceeds this threshold.
NFR-004	Shall	The system shall support at least 500 concurrent authenticated sessions per college/company tenant without degradation beyond the thresholds above.

5.2 Security

Req ID	Priority	Description
NFR-005	Shall	The system shall enforce Role-Based Access Control (RBAC) at the API gateway/middleware layer for every request.
NFR-006	Shall	The system shall authenticate all API requests using JSON Web Tokens (JWT) with a bounded expiry and refresh-token rotation.
NFR-007	Shall	The system shall support Multi-Factor Authentication (MFA) for College Admin, Company Admin, and Super Admin accounts.
NFR-008	Shall	The system shall hash all stored passwords using bcrypt with a cost factor reviewed at least annually.
NFR-009	Shall	The system shall enforce HTTPS/TLS for all client-server and server-server communication.
NFR-010	Shall	The system shall apply rate limiting to all public and authenticated API endpoints, backed by a Redis-based limiter store.
NFR-011	Shall	The system shall apply CAPTCHA challenges on registration and login endpoints upon detecting anomalous request volume.
NFR-012	Shall	The system shall scan all uploaded files (resumes, verification documents) with antivirus scanning prior to storage.
NFR-013	Shall	The system shall restrict uploaded file types and enforce a maximum file size per upload category.
NFR-014	Shall	The system shall restrict resume downloads to verified Recruiter accounts and shall log every download.

5.3 Reliability

Req ID	Priority	Description
NFR-015	Shall	The system shall isolate AI service calls in a dedicated module such that AI provider failures do not raise unhandled errors in core request paths.

Req ID	Priority	Description
NFR-016	Shall	The system shall implement automatic retry with exponential backoff for transient failures in background job processing.
NFR-017	Shall	The system shall maintain a dead-letter queue for background jobs that exhaust retry attempts, surfaced to the Super Admin for manual review.

5.4 Availability

Req ID	Priority	Description
NFR-018	Shall	The system shall target 99.5% monthly uptime for core (non-AI) platform functionality.
NFR-019	Shall	The system shall use cloud-based monitoring with automated alerting for service-down conditions.
NFR-020	Shall	The system shall support horizontal auto-scaling of backend service instances in response to load.

5.5 Scalability

Req ID	Priority	Description
NFR-021	Shall	The system's database schema shall support tenant partitioning strategies (e.g., indexed college_id/company_id) to sustain growth without cross-tenant query degradation.
NFR-022	Shall	The system shall support caching of frequently read, infrequently changed data (e.g., department lists, platform taxonomies) in Redis with explicit invalidation on write.

5.6 Maintainability

Req ID	Priority	Description
NFR-023	Shall	The system's codebase shall be organized into clearly bounded modules per domain (e.g., Coordinators, Listings, Applications, AI Services) with business logic isolated to the service layer.
NFR-024	Shall	The system shall maintain structured JSON logging with request/actor/tenant correlation identifiers on every mutating operation.
NFR-025	Should	The system shall maintain automated test coverage for all service-layer business logic, per the strategy in Section 10.1.

5.7 Usability

Req ID	Priority	Description
NFR-026	Shall	The system's web interface shall be responsive down to a 360px viewport width for all primary user-facing flows.
NFR-027	Shall	The system shall present field-level validation errors inline on forms, mapped from backend validation responses.
NFR-028	Should	The system shall provide empty-state guidance with a clear call-to-action on all primary list views.

5.8 Portability

Req ID	Priority	Description
NFR-029	Shall	The system's backend shall be containerized (Docker) to allow deployment across compatible cloud hosts without code changes.
NFR-030	Should	The system's frontend shall be deployable to any static/edge hosting provider supporting Next.js-equivalent React server rendering or a standard React SPA build.

5.9 Accessibility

Req ID	Priority	Description
NFR-031	Shall	The system shall conform to WCAG 2.1 Level AA for primary user-facing flows, including keyboard navigability and screen-reader-compatible labeling.
NFR-032	Shall	The system shall trap focus within modal dialogs and announce status changes via ARIA live regions.

6. Business Rules

6.1 Role-Based Rules

- **BR-001** A user account shall be assigned exactly one primary role at creation (College Admin, Placement Coordinator, College Student, Company Admin, Recruiter, External Student, or Super Admin).
- **BR-002** A Placement Coordinator's effective permissions shall be the intersection of their assigned permission set and their department scope.
- **BR-003** The Super Admin (PLATFORM_ADMIN) role shall never be conflated with, or granted the department-scoped permissions of, a Placement Coordinator.
- **BR-004** Cross-tenant data access shall be denied by default; any cross-tenant read (e.g., Super Admin analytics) shall be an explicit, audited exception.

6.2 Authentication Rules

- **BR-005** All accounts shall verify their email address before gaining full platform access.
- **BR-006** Invite tokens for coordinator and recruiter provisioning shall be single-use and shall expire within 72 hours of issuance.
- **BR-007** Accounts shall be locked for 15 minutes after 5 consecutive failed login attempts.
- **BR-008** College Admin, Company Admin, and Super Admin accounts shall be required to enable MFA within 14 days of first login.

6.3 College Rules

- **BR-009** A College Student account shall be associated with exactly one college tenant.
- **BR-010** A College Student shall only view and apply to listings approved by a Placement Coordinator of their college, or listings explicitly open to all colleges.
- **BR-011** A department shall not be deleted while referenced by an active student, coordinator, or listing record; it shall only be archived.

6.4 Company Rules

- **BR-012** A Company account shall remain in a Pending state, with listing-creation disabled, until GST/CIN and business email are verified and a Super Admin approves activation.
- **BR-013** A Recruiter account shall be scoped to exactly one company tenant and shall not post listings on behalf of another company.
- **BR-014** A listing created by a Recruiter shall be visible and editable by any Company Admin within the same company tenant.

6.5 External Student Rules

- **BR-015** An External Student account shall not be associated with any college tenant and shall not appear in any college's placement reporting.
- **BR-016** An External Student's access to premium features shall be strictly gated by active subscription status, verified server-side on every premium-endpoint request.

6.6 Super Admin Rules

- **BR-017** Every Super Admin mutating action shall be recorded in the audit log with actor identity, action, target, and before/after state where applicable.
- **BR-018** A Super Admin action that removes or suspends a tenant-level account (College, Company) shall require a documented reason.

6.7 Premium Feature Rules

- **BR-019** Premium features shall be strictly additive; no free-tier core workflow (search, apply, track application) shall be degraded or gated behind a subscription.
- **BR-020** Sponsored/prioritized placement of premium-tier listings in recommendation results shall be visibly labeled as such to the student.
- **BR-021** Subscription status shall be re-validated on each premium-feature request, not cached beyond a short TTL, to prevent stale-entitlement access after cancellation.

7. Data Requirements

7.1 Core Entities

The platform's data model follows a unified-entity strategy: Listing represents both Job and Internship (differentiated by a type enum) and Organization represents both College and Company (differentiated by a type enum), avoiding parallel duplicate schemas for structurally identical concepts.

- **User** — Base identity record for all account types; holds credentials, role, and status. Specialized by role via a discriminator.
- **Organization** — Unified model for a College or Company tenant, differentiated by a `type` enum, per the platform's unified-model architecture decision.
- **Listing** — Unified model for a Job or Internship posting, differentiated by a `type` enum, owned by a Company Organization and a posting Recruiter/Company Admin.
- **Department** — A College sub-unit used for student/coordinator scoping and reporting.
- **Application** — Represents a Student's application to a Listing, carrying a status lifecycle (Applied → Shortlisted → Interview → Selected/Rejected).
- **Resume** — A versioned resume document belonging to a Student, with parsed structured fields and an AI-derived score where available.
- **AuditLog** — Immutable record of a mutating action, capturing actor, action, target, tenant, and timestamp.
- **Notification** — An event-driven message queued for delivery to a User via one or more channels (email, in-app).
- **InviteToken** — A single-use, hashed, time-limited token used to activate Placement Coordinator and Recruiter accounts.
- **Subscription** — Represents an External Student's premium entitlement, including plan tier and validity window.
- **CompanyReview / SalaryInsight / InterviewExperience** — User-submitted research signals attached to a Company Organization, subject to moderation.

7.2 Entity Relationships

An Organization (type=COLLEGE) has many Users with role College Admin, Placement Coordinator, or College Student, and has many Departments. An Organization (type=COMPANY) has many Users with role Company Admin or Recruiter, and owns many Listings. A Listing belongs to exactly one Company Organization and one posting User (Recruiter or Company Admin), and has many Applications. An Application belongs to exactly one Student (College Student or External Student) and exactly one Listing, and progresses through the status lifecycle defined in Section 7.3. A Student has one active Resume with many historical Resume versions. Every mutating action across these entities produces one AuditLog record.

7.3 Data Dictionary

Field	Type	Values / Format	Description
Listing.type	ENUM	JOB INTERNSHIP	Differentiates a Listing without requiring separate tables.
Listing.status	ENUM	DRAFT PENDING_REVIEW APPROVED REJECTED ARCHIVED	Moderation lifecycle state.

Field	Type	Values / Format	Description
Organization.type	ENUM	COLLEGE COMPANY	Differentiates a tenant without requiring separate tables.
User.role	ENUM	COLLEGE_ADMIN PLACEMENT_COORDINATOR COLLEGE_STUDENT COMPANY_ADMIN RECRUITER EXTERNAL_STUDENT PLATFORM_ADMIN	Primary role discriminator.
Application.status	ENUM	APPLIED SHORTLISTED INTERVIEW OFFER_RELEASED SELECTED REJECTED	Application lifecycle state.
Company.verificationStatus	ENUM	PENDING VERIFIED REJECTED	Gates listing-creation capability.
Resume.version	INTEGER	Monotonically increasing per Student	Supports version history/restore.

7.4 Data Validation Rules

- **DV-001** All email fields shall be validated for RFC-5322 format both client-side and server-side.
- **DV-002** A Listing's application deadline shall be a future timestamp at creation time.
- **DV-003** A bulk student import row shall be rejected (with a reported reason) if the enrollment ID is missing, malformed, or a duplicate within the same college tenant.
- **DV-004** A company's GST/CIN shall be validated against the applicable format pattern before submission for Super Admin review.
- **DV-005** A resume upload shall be rejected if it exceeds the configured maximum file size or is not a PDF/DOCX file.

7.5 Data Retention Rules

- **DR-001** Audit logs shall be retained for a minimum of 3 years.
- **DR-002** Rejected/archived listings shall be retained in a soft-deleted state for reporting continuity; they shall not be hard-deleted.
- **DR-003** Free-tier resume version history shall retain the most recent 5 versions; premium-tier retention is extended per Section 3 (External Student, Premium Features).
- **DR-004** Invite tokens shall be purged 30 days after expiry or use.

7.6 Backup and Recovery

- **DB-001** The production database shall be backed up automatically on a daily schedule with point-in-time recovery enabled.
- **DB-002** Backup restore procedures shall be tested at least quarterly in a non-production environment.
- **DB-003** College-level Excel backup snapshots (Section 3, Placement Coordinator) are a user-facing convenience export and are not a substitute for the platform's database backup and recovery process.

8. Use Case Model

8.1 Actors

- College Admin
- Placement Coordinator
- College Student
- Company Admin
- Recruiter
- External Student
- Super Admin (PLATFORM_ADMIN)

8.2 High-Level Use Cases

- UC-01 Provision a Placement Coordinator or Recruiter account
- UC-02 Onboard students individually or in bulk
- UC-03 Submit a job or internship listing for review
- UC-04 Review, approve, or reject a listing
- UC-05 Search and apply to a listing
- UC-06 Track an application through to a hiring outcome
- UC-07 Generate and export placement reports
- UC-08 Analyze and score a resume
- UC-09 Approve a company's verification and activate the tenant
- UC-10 Investigate a moderation report and remove a listing

8.3 Level 2 Use Cases

Each high-level use case decomposes into role-specific sub-flows already enumerated as functional requirements in Chapter 3. For example, UC-04 (Review, Approve, or Reject a Listing) decomposes into the individual FR items under Section 3.1.2.1 (Listing Review & Moderation Workflow); UC-06 (Track an Application Through to a Hiring Outcome) spans FR items under Candidate Management, Interview Management, and the student-facing Application Tracking modules across Sections 3.1.3.4, 3.2.2.2, and 3.2.2.3. Section 8.5 provides the complete use-case-to-requirement traceability mapping.

8.4 Detailed Use Cases

UC-04 — Review, Approve, or Reject a Listing

Primary Actor: Placement Coordinator

Precondition: A Recruiter or Company Admin has submitted a Listing that is in PENDING_REVIEW status and is visible to the coordinator's college.

Postcondition: The Listing is in APPROVED status and visible to eligible students, or in REJECTED status with a reason recorded and the submitter notified.

Main Flow:

- 1. Coordinator opens the Listing Review queue, filtered to PENDING_REVIEW listings for their college.
- 2. Coordinator opens a listing and reviews its details, eligibility criteria, and company verification status.

- 3. Coordinator selects Approve or Reject.
- 4. If Reject, the system requires a reason before submission.
- 5. The system updates the Listing status, writes an audit log entry, and notifies the submitting Recruiter/Company Admin.
- 6. If Approve, the Listing becomes visible to eligible students and enters the recommendation pipeline.

Alternate Flow: *If the company's verification status is not VERIFIED, the system shall block approval and surface the verification gap to the coordinator.*

UC-05 – Search and Apply to a Listing

Primary Actor: College Student / External Student

Precondition: The student is authenticated and has a profile with at minimum an uploaded resume.

Postcondition: An Application record exists in APPLIED status, visible to the student and to the owning Recruiter/Company Admin.

Main Flow:

- 1. Student searches or filters listings by domain, location, salary/stipend, or work mode.
- 2. Student opens a listing and reviews eligibility criteria and deadline.
- 3. Student selects Apply.
- 4. The system validates eligibility (deadline not passed, criteria met) and creates an Application record.
- 5. The system notifies the owning Recruiter and, for College Students, the relevant Placement Coordinator.

Alternate Flow: *If the listing's deadline has passed between page load and submission, the system rejects the application with a clear expired-listing message.*

UC-09 – Approve a Company's Verification and Activate the Tenant

Primary Actor: Super Admin

Precondition: A Company Admin has submitted GST/CIN and a business email during onboarding; the tenant is in PENDING verification status.

Postcondition: The tenant is in VERIFIED status and listing-creation is enabled, or in REJECTED status with a reason recorded.

Main Flow:

- 1. Super Admin opens the pending-company verification queue.
- 2. Super Admin reviews submitted GST/CIN, business email domain match, and any supporting documents.
- 3. Super Admin approves or rejects the verification.
- 4. The system updates verificationStatus, writes an audit log entry, and notifies the Company Admin.

Alternate Flow: *If GST/CIN format validation fails automatically, the submission is blocked before reaching the Super Admin queue.*

8.5 Use Case Summary Table

Use Case	Title	Primary Actor(s)	Traceability
UC-01	Provision Coordinator/Recruiter	College Admin, Company Admin	FR set: Placement Coordinator Account Mgmt; Team & Recruiter Mgmt
UC-02	Onboard Students	College Admin	FR set: Student Account Management
UC-03	Submit Listing for Review	Recruiter, Company Admin	FR set: Job & Internship Posting/Listing

Use Case	Title	Primary Actor(s)	Traceability
			Management
UC-04	Review/Approve/Reject Listing	Placement Coordinator	FR set: Listing Review & Moderation Workflow
UC-05	Search and Apply	College Student, External Student	FR set: Search & Filters; Applications
UC-06	Track Application Outcome	Student, Recruiter, Coordinator	FR set: Application Tracking; Candidate Management; Interview Management
UC-07	Generate/Export Reports	College Admin, Placement Coordinator	FR set: Reports & Analytics Export; Data Export/Sync
UC-08	Analyze/Score Resume	Student	FR set: Resume Management Suite; Resume Intelligence
UC-09	Approve Company Verification	Super Admin	FR set: Platform Management
UC-10	Investigate Moderation Report	Super Admin	FR set: Moderation

9. System Design

9.1 Architecture Overview

IIB is built as a layered web application: a React.js client consumes a versioned REST API served by a Node.js/Express.js backend. Business logic is isolated in a service layer beneath thin Express controllers, following the pattern established in the College Module FDD (e.g., `CoordinatorsController` → `CoordinatorsService` → `CoordinatorsRepository` over Prisma). AI-dependent functionality is isolated into a dedicated AI services module that wraps the Google Gemini API, so that AI provider failures cannot propagate into core, non-AI request paths. Redis provides caching for read-heavy, infrequently changing data and backs rate-limiting and background-job queues.

9.2 Module Overview

- Identity & Access: authentication, JWT issuance, RBAC enforcement, MFA.
- College Module: College Admin, Placement Coordinator, College Student features (Section 3.1).
- Company Module: Company Admin, Recruiter features (Section 3.2).
- External Student Module: discovery, resume intelligence, AI career assistant, premium features (Section 3.3).
- Platform Module: Super Admin platform management, moderation, analytics, security, AI management (Section 3.4).
- AI Services Module: isolated Gemini integration serving recommendations, resume analysis, and generation features across all consuming modules.
- Notifications Module: event-driven email and in-app notification dispatch via background jobs.
- Data Export/Sync Module: Excel/CSV export and Google Sheets synchronization, treated strictly as derived output.

9.3 Technology Stack

Layer	Technology
Frontend	React.js (TypeScript)
Backend	Node.js + Express.js
ORM	Prisma
Database	MySQL
Cache / Queue	Redis
AI Provider	Google Gemini
Scraping (Phase 2, deferred)	Puppeteer / Cheerio / Playwright
Deployment	Docker (containerized) — or — Vercel (frontend) + Render (backend) + managed/Railway MySQL
Authentication	JWT with refresh-token rotation; MFA for privileged roles

9.4 Deployment Overview

The platform supports two equivalent deployment topologies: (a) fully containerized via Docker on any

compatible host, suited to team-managed infrastructure; or (b) a managed-platform topology with the frontend on Vercel, the backend on Render, and MySQL on a managed provider such as Railway, suited to rapid MVP delivery with minimal operations overhead. Both topologies share the same application code and environment-variable-driven configuration.

9.5 Integration Overview

- Google Gemini — AI recommendations, resume analysis/redesign, cover-letter generation, career roadmap.
- Google Sheets API — Placement Coordinator data synchronization.
- Email provider (SMTP/Resend-equivalent) — transactional email for invites and notifications.
- Antivirus scanning — uploaded file validation (resumes, verification documents).
- Payment/subscription provider — External Student premium feature entitlement (provider to be finalized; see Section 12.1).

10. Testing Requirements

10.1 Test Strategy

Testing follows a layered strategy: unit tests for service-layer business logic (target: all service methods implementing a SHALL requirement in Chapter 3), integration tests for API endpoints against a test database, and end-to-end tests for the primary flows in Section 8.5 (listing review, application, resume upload/scoring, company verification). AI-dependent features are tested against a mocked AI provider response and, separately, against the AI-unavailable fallback path required by Section 5.3.

10.2 Test Environment

- A dedicated staging environment shall mirror production topology (Section 9.4) at reduced scale.
- Nominal load for performance testing (Section 5.1) is defined as 500 concurrent authenticated sessions per tenant with a representative mix of read (80%) and write (20%) requests.
- Seed/fixture data shall cover at least one instance of every role, tenant type, and listing/application status value.

10.3 Test Case Template

Field	Description
Test Case ID	Unique identifier, e.g., TC-FR-004-01
Requirement ID	Traceable FR-/NFR-/BR- identifier under test
Preconditions	System/data state required before execution
Steps	Ordered actions to execute
Expected Result	Observable outcome that satisfies the requirement
Actual Result / Status	Pass / Fail / Blocked, recorded per test run

10.4 Acceptance Criteria

- Every Shall-priority FR/NFR in this document shall have at least one passing automated test before a release is tagged production-ready.
- No release shall ship with a known-open Security (Section 5.2) defect classified as High severity.
- AI-dependent features shall demonstrate correct fallback behavior with the AI provider disabled in the staging environment.

10.5 Validation Requirements

- Role-based access control shall be validated with negative test cases: every role shall be verified to receive 403 Forbidden on requests outside its permission set, not merely a hidden UI element.
- Tenant isolation shall be validated by attempting cross-tenant reads with a valid JWT from a different tenant and confirming denial.

11. Risks and Mitigation

The risks below are consolidated from the IIB Platform Risk Management Report and organized by category. Each risk carries two independent mitigations reflecting a defense-in-depth approach.

11.1 Technical Risks

Risk	Description	Mitigation – Primary	Mitigation – Secondary
Server Downtime	Platform becomes unavailable.	Use cloud monitoring and auto-scaling.	Maintain redundant backup servers.
Data Loss	Database failure or accidental deletion.	Perform automated daily backups.	Implement a disaster recovery plan.
API Abuse	Bots overload APIs using excessive requests.	Apply rate limiting.	Use CAPTCHA and API monitoring.

11.2 Data Risks

Risk	Description	Mitigation – Primary	Mitigation – Secondary
Fake Student Profiles	Students submit false information.	Verify email/mobile before account activation.	Allow colleges to verify enrolled students.
Spam/Fake Job Listings	Recruiters post misleading or duplicate jobs.	Use AI/manual moderation.	Enable user reporting and admin review.
Resume Theft	Unauthorized resume downloads.	Allow downloads only to verified recruiters.	Maintain resume download logs.

11.3 Security Risks

Risk	Description	Mitigation – Primary	Mitigation – Secondary
Unauthorized Access	Users access data beyond their role.	Implement Role-Based Access Control (RBAC).	Use JWT authentication and Multi-Factor Authentication (MFA).
Data Breach	Sensitive user data is exposed.	Encrypt passwords with bcrypt and enforce HTTPS.	Perform regular backups and secure the database.
Account Hacking	Accounts are compromised.	Enforce a strong password policy.	Enable MFA and login attempt limits.
Malicious File Upload	Users upload infected files.	Scan uploads with antivirus software.	Restrict file type and upload size.
Insider Threat	Admins misuse platform privileges.	Maintain detailed audit logs.	Follow the least-privilege access policy.

11.4 Business Risks

Risk	Description	Mitigation – Primary	Mitigation – Secondary
Fake Company	Fraudulent companies post	Verify GST/CIN and official business	Require Super Admin

Risk	Description	Mitigation – Primary	Mitigation – Secondary
Registration	fake jobs.	email.	approval before activation.

11.5 Mitigation Strategies – Cross-Cutting Summary

- Defense in depth: authentication (JWT + MFA), authorization (RBAC), and tenant scoping are enforced independently at multiple layers rather than relying on any single control.
- Verification before trust: both company registration (GST/CIN) and student enrollment are verified before an account gains write/apply capability, addressing fraud risk at the source rather than only through downstream moderation.
- Operational resilience: automated backups, disaster recovery, and cloud monitoring/auto-scaling collectively address availability and data-loss risk.
- Auditability: comprehensive audit, login, and access logging addresses insider-threat and post-incident investigation needs.

12. Future Enhancements

12.1 Planned Features

- Bulk CSV import for Placement Coordinator accounts, mirroring the existing Student bulk-import pattern.
- SSO/SAML provisioning for colleges and companies with an existing identity provider.
- Third-party company review integration (Glassdoor, AmbitionBox, Indeed) pending data-licensing agreements.

12.2 Optional Features

- Self-service coordinator/recruiter directory sync from an institution's or company's HR system via API.
- In-app real-time chat between Recruiter and applicant, beyond the current message/email/bulk-notification model.
- Employer-branded microsites generated from the Company profile and branding assets.

12.3 Phase 2 Scope

- Third-party job-board scraping (deferred past MVP per platform scope decisions) to supplement direct-posted listings.
- Predictive placement analytics (e.g., likelihood-to-place scoring) built on top of the existing AI services module.
- Mobile native applications for College Student and External Student roles.

13. Appendix

13.1 Glossary

- **College Admin** — Top-level administrator for a College tenant; provisions coordinators/students and configures college settings.
- **Placement Coordinator** — Department-scoped operational role reviewing listings and tracking placement outcomes on behalf of a college.
- **Company Admin** — Top-level administrator for a Company tenant; manages company profile, listings, and recruiter accounts.
- **Recruiter** — Company-scoped role managing individual job/internship postings and candidate pipelines.
- **External Student** — A student user not affiliated with any college tenant; self-registers and may access premium features.
- **Super Admin / PLATFORM_ADMIN** — Platform operations/developer role with cross-tenant visibility, distinct from any placement-adjacent function.
- **Listing** — Unified data model representing either a Job or an Internship posting, differentiated by a type field.
- **Organization** — Unified data model representing either a College or a Company tenant, differentiated by a type field.
- **Tenant** — An isolated College or Company account whose data is scoped and never visible to other tenants by default.
- **RBAC** — Role-Based Access Control — authorization model restricting actions based on assigned role and permissions.
- **FDD** — Feature Design Document — a production-level, 20-section specification for a single feature, more granular than this SRS.

13.2 Abbreviations

Abbreviation	Full Form
IIB	Industry Intelligence Bridge
SRS	Software Requirements Specification
FDD	Feature Design Document
FR	Functional Requirement
NFR	Non-Functional Requirement
BR	Business Rule
UC	Use Case
RBAC	Role-Based Access Control
JWT	JSON Web Token
MFA	Multi-Factor Authentication
ORM	Object-Relational Mapping
API	Application Programming Interface

Abbreviation	Full Form
REST	Representational State Transfer
ATS	Applicant Tracking System
GST/CIN	Goods and Services Tax Identification / Corporate Identification Number (India company verification identifiers)
PPO	Pre-Placement Offer
LOR	Letter of Recommendation
WCAG	Web Content Accessibility Guidelines
MVP	Minimum Viable Product

13.3 Requirement Index

This index lists all 250 functional and non-functional requirements defined in this document, in issuance order, with their source location for traceability.

ID	Priority	Description	Location
FR-001	Shall	The system shall allow a College Admin to create a Placement Coordinator account with name, email, and one or more department assignments.	§3.1.1.1 Placement Coordinator Account Management
FR-002	Shall	The system shall send a single-use, time-limited (72-hour) email invite token to a newly created Placement Coordinator to activate their account.	§3.1.1.1 Placement Coordinator Account Management
FR-003	Shall	The system shall allow a College Admin to edit a Placement Coordinator's name and department assignments.	§3.1.1.1 Placement Coordinator Account Management
FR-004	Shall	The system shall allow a College Admin to suspend or reactivate a Placement Coordinator account.	§3.1.1.1 Placement Coordinator Account Management
FR-005	Shall	The system shall allow a College Admin to remove (soft-delete) a Placement Coordinator account, blocking removal if the coordinator has pending moderation actions assigned.	§3.1.1.1 Placement Coordinator Account Management
FR-006	Shall	The system shall allow a College Admin to resend an invite, invalidating any prior unused invite token.	§3.1.1.1 Placement Coordinator Account Management
FR-007	Shall	The system shall reject duplicate Placement Coordinator email addresses within the same college tenant.	§3.1.1.1 Placement Coordinator Account Management
FR-008	Shall	The system shall allow a College Admin to create an individual College Student account with name, email, enrollment ID, and department.	§3.1.1.2 Student Account Management (Individual & Bulk)
FR-009	Shall	The system shall allow a College Admin to bulk-import College Student accounts via a CSV/Excel upload, validating each row before commit.	§3.1.1.2 Student Account Management (Individual & Bulk)
FR-010	Shall	The system shall report row-level validation errors from a bulk import without discarding valid rows.	§3.1.1.2 Student Account Management (Individual & Bulk)
FR-011	Shall	The system shall allow a College Admin to edit a student's department, batch/year, and contact details.	§3.1.1.2 Student Account Management (Individual & Bulk)
FR-012	Shall	The system shall allow a College Admin to block or remove a student account.	§3.1.1.2 Student Account Management (Individual & Bulk)
FR-013	Shall	The system shall allow colleges to mark a student as verified/enrolled to distinguish genuine students from unverified self-registrations.	§3.1.1.2 Student Account Management (Individual & Bulk)

ID	Priority	Description	Location
FR-014	Shall	The system shall allow a College Admin to create, rename, and archive departments.	§3.1.1.3 Department Management
FR-015	Shall	The system shall prevent hard deletion of a department that is referenced by any student, coordinator, or listing record; departments in use shall only be archivable.	§3.1.1.3 Department Management
FR-016	Shall	The system shall allow a College Admin to view a count of students and coordinators per department.	§3.1.1.3 Department Management
FR-017	Shall	The system shall allow a College Admin to edit the college profile, including name, address, accreditation details, and website.	§3.1.1.4 College Profile, Branding & Platform Settings
FR-018	Shall	The system shall allow a College Admin to upload a college logo and set brand colors used on college-facing pages.	§3.1.1.4 College Profile, Branding & Platform Settings
FR-019	Shall	The system shall allow a College Admin to configure college-level platform settings, including default application deadlines and listing visibility windows.	§3.1.1.4 College Profile, Branding & Platform Settings
FR-020	Shall	The system shall allow a College Admin to assign granular permissions (e.g., approve-listings, export-data, manage-students) to a Placement Coordinator independent of department scope.	§3.1.1.5 Roles, Permissions & Access Control
FR-021	Shall	The system shall enforce that every College Admin and Placement Coordinator action is authorized against the caller's role and permission set on every request, not only at the UI layer.	§3.1.1.5 Roles, Permissions & Access Control
FR-022	Shall	The system shall allow a College Admin to view an access-control summary listing each coordinator's effective permissions.	§3.1.1.5 Roles, Permissions & Access Control
FR-023	Shall	The system shall allow a College Admin to export a department-wise placement report in Excel (.xlsx) format.	§3.1.1.6 Reports & Analytics Export
FR-024	Shall	The system shall allow a College Admin to export placement data in CSV format.	§3.1.1.6 Reports & Analytics Export
FR-025	Shall	The system shall allow a College Admin to generate a monthly placement summary report.	§3.1.1.6 Reports & Analytics Export
FR-026	Shall	The system shall allow a College Admin to generate an annual placement summary report.	§3.1.1.6 Reports & Analytics Export
FR-027	Shall	The system shall generate all exports from live database records, treating Excel/CSV output strictly as a derived artifact, never as a system of record.	§3.1.1.6 Reports & Analytics Export
FR-028	Shall	The system shall allow a College Admin to enable or disable college-level notification categories (e.g., student-selected, internship-started).	§3.1.1.7 Notification Settings & Engagement Triggers
FR-029	Shall	The system shall notify a College Admin when a student under their college is marked as Selected by a company.	§3.1.1.7 Notification Settings & Engagement Triggers
FR-030	Shall	The system shall notify a College Admin when a student under their college begins an internship.	§3.1.1.7 Notification Settings & Engagement Triggers
FR-031	Should	The system shall allow a College Admin to configure the notification channels (email, in-app) used for each category.	§3.1.1.7 Notification Settings & Engagement Triggers
FR-032	Shall	The system shall present newly submitted job listings to Placement Coordinators for review before publication to students.	§3.1.2.1 Listing Review & Moderation Workflow
FR-033	Shall	The system shall present newly submitted internship listings to Placement Coordinators for review before publication to students.	§3.1.2.1 Listing Review & Moderation Workflow
FR-034	Shall	The system shall allow a Placement Coordinator to approve a listing, making it visible to eligible students.	§3.1.2.1 Listing Review & Moderation Workflow
FR-035	Shall	The system shall allow a Placement Coordinator to reject a listing with a mandatory reason, notifying the submitting	§3.1.2.1 Listing Review & Moderation Workflow

ID	Priority	Description	Location
		Recruiter/Company Admin.	
FR-036	Shall	The system shall allow a Placement Coordinator to edit listing metadata (deadline, eligibility criteria) prior to approval.	§3.1.2.1 Listing Review & Moderation Workflow
FR-037	Shall	The system shall allow a Placement Coordinator to archive an expired or withdrawn listing.	§3.1.2.1 Listing Review & Moderation Workflow
FR-038	Should	The system shall allow a Placement Coordinator to pin a listing to the top of the student-facing feed.	§3.1.2.1 Listing Review & Moderation Workflow
FR-039	Shall	The system shall allow a Placement Coordinator to export current placement data to Excel on demand.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-040	Shall	The system shall allow a Placement Coordinator to export current placement data to Google Sheets on demand.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-041	Shall	The system shall support a scheduled auto-sync of placement data to a connected Excel file at a coordinator-configured interval.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-042	Shall	The system shall support a scheduled auto-sync of placement data to a connected Google Sheet at a coordinator-configured interval.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-043	Shall	The system shall allow a Placement Coordinator to create a manual backup snapshot of college placement data in Excel format.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-044	Shall	The system shall allow a Placement Coordinator to restore college placement data from a prior backup snapshot, with a confirmation step and audit log entry.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-045	Shall	The system shall never treat an imported Excel or Google Sheets file as authoritative; any conflicting inbound edits shall require explicit coordinator confirmation before being applied to the database.	§3.1.2.2 Data Export, Google Sheets/Excel Auto-Sync & Backup-Restore
FR-046	Shall	The system shall allow a Placement Coordinator to generate a Letter of Recommendation for a student from a standard template pre-filled with student and placement data.	§3.1.2.3 Letter of Recommendation Generator (LOR)
FR-047	Shall	The system shall allow a Placement Coordinator to edit the generated LOR content before finalizing it.	§3.1.2.3 Letter of Recommendation Generator (LOR)
FR-048	Shall	The system shall allow a Placement Coordinator to export a finalized LOR as a signed PDF document.	§3.1.2.3 Letter of Recommendation Generator (LOR)
FR-049	Should	The system shall maintain a history of LORs issued per student for audit purposes.	§3.1.2.3 Letter of Recommendation Generator (LOR)
FR-050	Shall	The system shall allow a Placement Coordinator to view the list of students within their assigned department(s).	§3.1.2.4 Student & Application Tracking Console
FR-051	Shall	The system shall allow a Placement Coordinator to track the application status of any student under their scope, from applied through to selected/rejected.	§3.1.2.4 Student & Application Tracking Console
FR-052	Shall	The system shall allow a Placement Coordinator to view a student's overall placement status.	§3.1.2.4 Student & Application Tracking Console
FR-053	Shall	The system shall allow a Placement Coordinator to view a student's internship status.	§3.1.2.4 Student & Application Tracking Console
FR-054	Shall	The system shall allow a Placement Coordinator to view a student's uploaded resume, subject to resume-access logging.	§3.1.2.4 Student & Application Tracking Console

ID	Priority	Description	Location
FR-055	Shall	The system shall allow a Placement Coordinator to filter the student list by department.	§3.1.2.4 Student & Application Tracking Console
FR-056	Shall	The system shall notify a Placement Coordinator when a student under their scope applies to a listing.	§3.1.2.5 Coordinator Notification Hub
FR-057	Shall	The system shall notify a Placement Coordinator when a student is shortlisted.	§3.1.2.5 Coordinator Notification Hub
FR-058	Shall	The system shall notify a Placement Coordinator when a student is selected.	§3.1.2.5 Coordinator Notification Hub
FR-059	Shall	The system shall notify a Placement Coordinator when a student's internship starts.	§3.1.2.5 Coordinator Notification Hub
FR-060	Shall	The system shall notify a Placement Coordinator when a student's internship is completed.	§3.1.2.5 Coordinator Notification Hub
FR-061	Shall	The system shall notify a Placement Coordinator when a student receives a Pre-Placement Offer (PPO).	§3.1.2.5 Coordinator Notification Hub
FR-062	Shall	The system shall notify a Placement Coordinator when a student accepts a job offer.	§3.1.2.5 Coordinator Notification Hub
FR-063	Shall	The system shall display the count of students placed, scoped by department and time period.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-064	Shall	The system shall display company hiring trends for the coordinator's college.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-065	Shall	The system shall display department-wise placement performance.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-066	Shall	The system shall display the highest package offered within a selected period.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-067	Shall	The system shall display the average package offered within a selected period.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-068	Shall	The system shall display the count of currently active opportunities visible to the college.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-069	Shall	The system shall allow a Placement Coordinator to search for a student by name, enrollment ID, or department.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-070	Shall	The system shall allow a Placement Coordinator to search for a company by name.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-071	Shall	The system shall allow a Placement Coordinator to search for an internship listing.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-072	Shall	The system shall allow a Placement Coordinator to search for a job listing.	§3.1.2.6 Placement Analytics Dashboard & Unified Search
FR-073	Shall	The system shall display a personalized dashboard summarizing recommended opportunities, saved items, and application status for the logged-in student.	§3.1.3.1 Personalized Dashboard & AI Recommendations
FR-074	Shall	The system shall display AI-recommended job listings based on the student's profile and resume, with a rule-based fallback ranking if the AI service is unavailable.	§3.1.3.1 Personalized Dashboard & AI Recommendations
FR-075	Shall	The system shall display AI-recommended internship listings based on the student's profile and resume, with a rule-based fallback ranking if the AI service is unavailable.	§3.1.3.1 Personalized Dashboard & AI Recommendations
FR-076	Shall	The system shall display recently added listings relevant to the student's department and eligibility.	§3.1.3.1 Personalized Dashboard & AI Recommendations
FR-077	Shall	The system shall display listings closing within a configurable threshold (e.g., 72 hours).	§3.1.3.1 Personalized Dashboard & AI Recommendations

ID	Priority	Description	Location
FR-078	Shall	The system shall allow a College Student to search job listings by keyword.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-079	Shall	The system shall allow a College Student to search internship listings by keyword.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-080	Shall	The system shall allow a College Student to search companies by name.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-081	Shall	The system shall allow a College Student to filter listings by domain/field.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-082	Shall	The system shall allow a College Student to filter listings by location.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-083	Shall	The system shall allow a College Student to filter job listings by salary range.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-084	Shall	The system shall allow a College Student to filter internship listings by stipend range.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-085	Shall	The system shall allow a College Student to filter listings by required experience level.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-086	Shall	The system shall allow a College Student to filter listings by work mode (remote/hybrid/on-site).	§3.1.3.2 Job/Internship/Company Search & Filters
FR-087	Shall	The system shall allow a College Student to filter listings by company.	§3.1.3.2 Job/Internship/Company Search & Filters
FR-088	Shall	The system shall allow a College Student to upload a resume in PDF or DOCX format.	§3.1.3.3 Resume Management Suite
FR-089	Shall	The system shall provide an AI-generated resume analysis summarizing strengths and gaps, degrading gracefully to a checklist-based analysis if the AI service is unavailable.	§3.1.3.3 Resume Management Suite
FR-090	Shall	The system shall compute and display a resume score based on completeness and formatting heuristics.	§3.1.3.3 Resume Management Suite
FR-091	Should	The system shall provide a set of resume templates the student may apply to a builder-generated resume.	§3.1.3.3 Resume Management Suite
FR-092	Shall	The system shall provide a guided resume builder that produces a downloadable resume document.	§3.1.3.3 Resume Management Suite
FR-093	Shall	The system shall retain prior resume versions and allow a student to restore or compare versions.	§3.1.3.3 Resume Management Suite
FR-094	Shall	The system shall allow a College Student to apply to an approved, eligible job or internship listing.	§3.1.3.4 Apply / Save / Bookmark & Application Tracking
FR-095	Shall	The system shall allow a College Student to save a listing for later review without applying.	§3.1.3.4 Apply / Save / Bookmark & Application Tracking
FR-096	Shall	The system shall allow a College Student to bookmark a listing.	§3.1.3.4 Apply / Save / Bookmark & Application Tracking
FR-097	Shall	The system shall allow a College Student to track the status of each submitted application.	§3.1.3.4 Apply / Save / Bookmark & Application Tracking
FR-098	Shall	The system shall display a chronological application history for the student.	§3.1.3.4 Apply / Save / Bookmark & Application Tracking
FR-099	Shall	The system shall allow a College Student to list and edit skills on their profile.	§3.1.3.5 Student Profile
FR-100	Shall	The system shall allow a College Student to enter education details, pre-populated from college enrollment data where available.	§3.1.3.5 Student Profile
FR-101	Shall	The system shall allow a College Student to list projects with	§3.1.3.5 Student Profile

ID	Priority	Description	Location
		descriptions and links.	
FR-102	Shall	The system shall allow a College Student to list certificates with issuing authority and date.	§3.1.3.5 Student Profile
FR-103	Should	The system shall allow a College Student to list career interests used to inform recommendations.	§3.1.3.5 Student Profile
FR-104	Shall	The system shall display aggregated company reviews submitted by verified platform users.	§3.1.3.6 Company Reviews, Salary Insights & Interview Experience
FR-105	Shall	The system shall display salary insights for a company or role where sufficient data exists.	§3.1.3.6 Company Reviews, Salary Insights & Interview Experience
FR-106	Shall	The system shall allow a College Student to view and submit interview experience write-ups tied to a specific company and role.	§3.1.3.6 Company Reviews, Salary Insights & Interview Experience
FR-107	Shall	The system shall allow a Company Admin to create and edit the company profile, including industry, size, and description.	§3.2.1.1 Company Management
FR-108	Shall	The system shall allow a Company Admin to manage one or more branch/office locations under the company profile.	§3.2.1.1 Company Management
FR-109	Shall	The system shall require a Company Admin to submit GST/CIN and an official business email during onboarding, with the account held in a pending state until Super Admin approval.	§3.2.1.1 Company Management
FR-110	Shall	The system shall allow a Company Admin to upload company branding assets (logo, cover image) for the public company profile.	§3.2.1.1 Company Management
FR-111	Shall	The system shall allow a Company Admin to create job listings.	§3.2.1.2 Job & Internship Listing Management
FR-112	Shall	The system shall allow a Company Admin to create internship listings.	§3.2.1.2 Job & Internship Listing Management
FR-113	Shall	The system shall allow a Company Admin to edit any listing owned by the company.	§3.2.1.2 Job & Internship Listing Management
FR-114	Shall	The system shall allow a Company Admin to delete a listing owned by the company.	§3.2.1.2 Job & Internship Listing Management
FR-115	Shall	The system shall allow a Company Admin to archive an expired listing owned by the company.	§3.2.1.2 Job & Internship Listing Management
FR-116	Shall	The system shall display view counts for each company listing.	§3.2.1.3 Hiring Analytics & Reports
FR-117	Shall	The system shall display the count of applicants per listing.	§3.2.1.3 Hiring Analytics & Reports
FR-118	Shall	The system shall display the count of shortlisted candidates per listing.	§3.2.1.3 Hiring Analytics & Reports
FR-119	Shall	The system shall display the count of selected candidates per listing.	§3.2.1.3 Hiring Analytics & Reports
FR-120	Shall	The system shall allow a Company Admin to export a consolidated hiring report.	§3.2.1.3 Hiring Analytics & Reports
FR-121	Shall	The system shall allow a Company Admin to add a Recruiter account, scoped to the company tenant.	§3.2.1.4 Team & Recruiter Management
FR-122	Shall	The system shall allow a Company Admin to remove a Recruiter account.	§3.2.1.4 Team & Recruiter Management
FR-123	Shall	The system shall allow a Company Admin to assign granular permissions to a Recruiter (e.g., post-listing, view-applicants only).	§3.2.1.4 Team & Recruiter Management

ID	Priority	Description	Location
FR-124	Shall	The system shall allow a Recruiter to post a new internship listing, subject to college/placement-coordinator review where the listing targets a specific college.	§3.2.2.1 Job & Internship Posting Management
FR-125	Shall	The system shall allow a Recruiter to post a new job listing.	§3.2.2.1 Job & Internship Posting Management
FR-126	Shall	The system shall allow a Recruiter to edit their own listing prior to its deadline.	§3.2.2.1 Job & Internship Posting Management
FR-127	Shall	The system shall allow a Recruiter to close a listing before its scheduled deadline.	§3.2.2.1 Job & Internship Posting Management
FR-128	Shall	The system shall allow a Recruiter to extend a listing's application deadline.	§3.2.2.1 Job & Internship Posting Management
FR-129	Shall	The system shall allow a Recruiter to view all applicants to their listing.	§3.2.2.2 Candidate Management
FR-130	Shall	The system shall allow a Recruiter to filter applicants by criteria such as college, department, or resume score.	§3.2.2.2 Candidate Management
FR-131	Shall	The system shall allow a Recruiter to download an applicant's resume, and shall log every resume download with recruiter identity and timestamp.	§3.2.2.2 Candidate Management
FR-132	Shall	The system shall allow a Recruiter to shortlist a candidate.	§3.2.2.2 Candidate Management
FR-133	Shall	The system shall allow a Recruiter to reject a candidate.	§3.2.2.2 Candidate Management
FR-134	Shall	The system shall allow a Recruiter to schedule an interview for a shortlisted candidate.	§3.2.2.3 Interview Management
FR-135	Shall	The system shall allow a Recruiter to update the status of a scheduled interview.	§3.2.2.3 Interview Management
FR-136	Shall	The system shall allow a Recruiter to mark a candidate as Selected.	§3.2.2.3 Interview Management
FR-137	Shall	The system shall allow a Recruiter to mark a candidate as Rejected.	§3.2.2.3 Interview Management
FR-138	Shall	The system shall allow a Recruiter to mark an offer as Released for a selected candidate.	§3.2.2.3 Interview Management
FR-139	Shall	The system shall allow a Recruiter to send an in-platform message to an applicant.	§3.2.2.4 Communication
FR-140	Shall	The system shall allow a Recruiter to email a candidate directly from the platform.	§3.2.2.4 Communication
FR-141	Shall	The system shall allow a Recruiter to send a bulk notification to all applicants of a given listing.	§3.2.2.4 Communication
FR-142	Shall	The system shall display AI-recommended job listings on the External Student dashboard, with a rule-based fallback if the AI service is unavailable.	§3.3.1.1 Dashboard & Discovery
FR-143	Shall	The system shall display AI-recommended internship listings on the External Student dashboard, with a rule-based fallback if the AI service is unavailable.	§3.3.1.1 Dashboard & Discovery
FR-144	Shall	The system shall display trending job listings based on platform-wide application velocity.	§3.3.1.1 Dashboard & Discovery
FR-145	Shall	The system shall display trending internship listings based on platform-wide application velocity.	§3.3.1.1 Dashboard & Discovery
FR-146	Shall	The system shall display newly published opportunities matching the student's stated interests.	§3.3.1.1 Dashboard & Discovery
FR-147	Shall	The system shall allow an External Student to search job listings by keyword.	§3.3.1.2 Search & Smart Filters

ID	Priority	Description	Location
FR-148	Shall	The system shall allow an External Student to search internship listings by keyword.	§3.3.1.2 Search & Smart Filters
FR-149	Shall	The system shall allow an External Student to search companies by name.	§3.3.1.2 Search & Smart Filters
FR-150	Shall	The system shall allow an External Student to filter listings by required skill.	§3.3.1.2 Search & Smart Filters
FR-151	Shall	The system shall allow an External Student to filter listings by domain.	§3.3.1.2 Search & Smart Filters
FR-152	Shall	The system shall allow an External Student to filter job listings by salary range.	§3.3.1.2 Search & Smart Filters
FR-153	Shall	The system shall allow an External Student to filter internship listings by stipend range.	§3.3.1.2 Search & Smart Filters
FR-154	Shall	The system shall allow an External Student to filter listings restricted to fully remote work.	§3.3.1.2 Search & Smart Filters
FR-155	Shall	The system shall allow an External Student to filter listings restricted to hybrid work.	§3.3.1.2 Search & Smart Filters
FR-156	Shall	The system shall allow an External Student to filter listings by required experience level.	§3.3.1.2 Search & Smart Filters
FR-157	Shall	The system shall allow an External Student to filter listings by city.	§3.3.1.2 Search & Smart Filters
FR-158	Shall	The system shall allow an External Student to filter listings by company.	§3.3.1.2 Search & Smart Filters
FR-159	Shall	The system shall allow an External Student to upload a resume in PDF or DOCX format.	§3.3.1.3 Resume Intelligence
FR-160	Shall	The system shall parse an uploaded resume into structured fields (skills, education, experience).	§3.3.1.3 Resume Intelligence
FR-161	Shall	The system shall compute a resume score based on completeness and formatting heuristics.	§3.3.1.3 Resume Intelligence
FR-162	Shall	The system shall provide an AI-generated resume redesign suggestion, degrading gracefully to a static best-practices checklist if the AI service is unavailable.	§3.3.1.3 Resume Intelligence
FR-163	Shall	The system shall provide a guided resume builder that produces a downloadable resume document.	§3.3.1.3 Resume Intelligence
FR-164	Should	The system shall provide a set of resume templates for use in the resume builder.	§3.3.1.3 Resume Intelligence
FR-165	Shall	The system shall provide an ATS-compatibility check that flags formatting likely to fail automated parsing.	§3.3.1.3 Resume Intelligence
FR-166	Shall	The system shall provide AI-generated job recommendations based on the student's profile and resume.	§3.3.1.4 AI Career Assistant
FR-167	Shall	The system shall provide AI-generated internship recommendations based on the student's profile and resume.	§3.3.1.4 AI Career Assistant
FR-168	Shall	The system shall provide an AI-generated skill-gap analysis comparing the student's profile against target roles.	§3.3.1.4 AI Career Assistant
FR-169	Should	The system shall provide AI-generated learning-resource suggestions to close identified skill gaps.	§3.3.1.4 AI Career Assistant
FR-170	Shall	The system shall provide an AI-generated career roadmap outlining suggested next steps.	§3.3.1.4 AI Career Assistant
FR-171	Shall	The system shall provide an AI-generated cover letter drafted for a specific listing, editable before submission.	§3.3.1.4 AI Career Assistant

ID	Priority	Description	Location
FR-172	Shall	The system shall display aggregated company reviews submitted by verified platform users.	§3.3.1.5 Company Insights
FR-173	Should	The system shall display third-party company review signals (e.g., Glassdoor) where a licensed data source is integrated; this is deferred past MVP pending third-party licensing.	§3.3.1.5 Company Insights
FR-174	Should	The system shall display third-party company review signals (e.g., AmbitionBox) where a licensed data source is integrated; this is deferred past MVP pending third-party licensing.	§3.3.1.5 Company Insights
FR-175	Should	The system shall display third-party company review signals (e.g., Indeed) where a licensed data source is integrated; this is deferred past MVP pending third-party licensing.	§3.3.1.5 Company Insights
FR-176	Shall	The system shall display salary insights for a company or role where sufficient platform data exists.	§3.3.1.5 Company Insights
FR-177	Shall	The system shall display hiring trend data for a company over a selected period.	§3.3.1.5 Company Insights
FR-178	Shall	The system shall allow an External Student to apply to an approved, eligible listing.	§3.3.1.6 Applications
FR-179	Shall	The system shall allow an External Student to save a listing without applying.	§3.3.1.6 Applications
FR-180	Shall	The system shall allow an External Student to bookmark a listing.	§3.3.1.6 Applications
FR-181	Shall	The system shall allow an External Student to track application status.	§3.3.1.6 Applications
FR-182	Shall	The system shall display an application timeline showing status changes with timestamps.	§3.3.1.6 Applications
FR-183	Shall	The system shall provide enhanced AI resume optimization to subscribed premium students.	§3.3.1.7 Premium Features
FR-184	Shall	The system shall provide an AI career coach conversational feature to subscribed premium students.	§3.3.1.7 Premium Features
FR-185	Shall	The system shall provide unlimited job/internship alerts to subscribed premium students, versus a capped alert count for free-tier students.	§3.3.1.7 Premium Features
FR-186	Shall	The system shall prioritize premium students' listings in AI recommendation ranking, clearly labeled as sponsored placement where applicable.	§3.3.1.7 Premium Features
FR-187	Shall	The system shall provide advanced analytics (e.g., application-to-interview conversion rate) to subscribed premium students.	§3.3.1.7 Premium Features
FR-188	Should	The system shall provide an extended resume version history to subscribed premium students beyond the free-tier retention limit.	§3.3.1.7 Premium Features
FR-189	Shall	The system shall allow a Super Admin to view, approve, suspend, or remove College tenant accounts.	§3.4.1.1 Platform Management
FR-190	Shall	The system shall allow a Super Admin to view, approve, suspend, or remove Company tenant accounts, including final approval of company GST/CIN verification.	§3.4.1.1 Platform Management
FR-191	Shall	The system shall allow a Super Admin to view and, where necessary, suspend Student accounts across all tenants.	§3.4.1.1 Platform Management
FR-192	Shall	The system shall allow a Super Admin to view and, where necessary, suspend Recruiter accounts across all tenants.	§3.4.1.1 Platform Management
FR-193	Shall	The system shall allow a Super Admin to create and manage other platform-level admin accounts, with all such actions captured in the	§3.4.1.1 Platform Management

ID	Priority	Description	Location
		audit log.	
FR-194	Should	The system shall allow a Super Admin to configure and enable a scraper-based data source; this capability is deferred past MVP.	§3.4.1.2 Data Sources
FR-195	Should	The system shall allow a Super Admin to configure a third-party API integration as a listing source; this capability is deferred past MVP.	§3.4.1.2 Data Sources
FR-196	Should	The system shall allow a Super Admin to register a new external platform as a candidate data source; this capability is deferred past MVP.	§3.4.1.2 Data Sources
FR-197	Shall	The system shall provide the Super Admin with monitoring of configured data source health and last-successful-sync status.	§3.4.1.2 Data Sources
FR-198	Shall	The system shall allow a Super Admin to review listings and profiles flagged/reported by users.	§3.4.1.3 Moderation
FR-199	Shall	The system shall allow a Super Admin to remove a listing identified as spam or fraudulent.	§3.4.1.3 Moderation
FR-200	Shall	The system shall allow a Super Admin to manage the platform's listing category taxonomy.	§3.4.1.3 Moderation
FR-201	Shall	The system shall allow a Super Admin to manage the platform's domain/industry taxonomy.	§3.4.1.3 Moderation
FR-202	Shall	The system shall display platform-wide growth metrics over a selected period.	§3.4.1.4 Analytics
FR-203	Shall	The system shall display user growth metrics segmented by role.	§3.4.1.4 Analytics
FR-204	Shall	The system shall display the count of currently active job listings platform-wide.	§3.4.1.4 Analytics
FR-205	Shall	The system shall display the count of currently active internship listings platform-wide.	§3.4.1.4 Analytics
FR-206	Shall	The system shall display company growth metrics over a selected period.	§3.4.1.4 Analytics
FR-207	Shall	The system shall display college growth metrics over a selected period.	§3.4.1.4 Analytics
FR-208	Shall	The system shall display a revenue dashboard summarizing premium subscription and any other monetized activity.	§3.4.1.4 Analytics
FR-209	Shall	The system shall provide the Super Admin with a searchable audit log of all mutating actions platform-wide, including actor, action, target, and timestamp.	§3.4.1.5 Security
FR-210	Shall	The system shall provide the Super Admin with a login log capturing successful and failed authentication attempts.	§3.4.1.5 Security
FR-211	Shall	The system shall provide the Super Admin with an access log for sensitive resource reads (e.g., resume downloads).	§3.4.1.5 Security
FR-212	Shall	The system shall provide the Super Admin with API request monitoring, including error-rate and latency visibility.	§3.4.1.5 Security
FR-213	Shall	The system shall provide the Super Admin with backup management, including manual trigger and restore-point selection.	§3.4.1.5 Security
FR-214	Shall	The system shall allow a Super Admin to configure the active AI model/provider used by the AI services module.	§3.4.1.6 AI Management
FR-215	Shall	The system shall allow a Super Admin to manage and version prompt templates used by AI features.	§3.4.1.6 AI Management
FR-216	Shall	The system shall allow a Super Admin to monitor recommendation-engine performance and override ranking weights.	§3.4.1.6 AI Management

ID	Priority	Description	Location
FR-217	Shall	The system shall allow a Super Admin to monitor resume-engine (parsing/scoring) performance and accuracy.	§3.4.1.6 AI Management
FR-218	Shall	The system shall isolate all AI service calls in a dedicated module such that an AI provider outage does not block core, non-AI platform workflows.	§3.4.1.6 AI Management
NFR-001	Shall	The system shall return search and listing API responses within 500ms at the 95th percentile under nominal load (defined in Section 10.2).	§5 5.1 Performance
NFR-002	Shall	The system shall complete an Excel/CSV export of up to 10,000 records within 15 seconds.	§5 5.1 Performance
NFR-003	Shall	AI-augmented endpoints (recommendations, resume analysis) shall return within 5 seconds at the 95th percentile, or fall back to a non-AI default response within 2 seconds if the AI provider exceeds this threshold.	§5 5.1 Performance
NFR-004	Shall	The system shall support at least 500 concurrent authenticated sessions per college/company tenant without degradation beyond the thresholds above.	§5 5.1 Performance
NFR-005	Shall	The system shall enforce Role-Based Access Control (RBAC) at the API gateway/middleware layer for every request.	§5 5.2 Security
NFR-006	Shall	The system shall authenticate all API requests using JSON Web Tokens (JWT) with a bounded expiry and refresh-token rotation.	§5 5.2 Security
NFR-007	Shall	The system shall support Multi-Factor Authentication (MFA) for College Admin, Company Admin, and Super Admin accounts.	§5 5.2 Security
NFR-008	Shall	The system shall hash all stored passwords using bcrypt with a cost factor reviewed at least annually.	§5 5.2 Security
NFR-009	Shall	The system shall enforce HTTPS/TLS for all client-server and server-server communication.	§5 5.2 Security
NFR-010	Shall	The system shall apply rate limiting to all public and authenticated API endpoints, backed by a Redis-based limiter store.	§5 5.2 Security
NFR-011	Shall	The system shall apply CAPTCHA challenges on registration and login endpoints upon detecting anomalous request volume.	§5 5.2 Security
NFR-012	Shall	The system shall scan all uploaded files (resumes, verification documents) with antivirus scanning prior to storage.	§5 5.2 Security
NFR-013	Shall	The system shall restrict uploaded file types and enforce a maximum file size per upload category.	§5 5.2 Security
NFR-014	Shall	The system shall restrict resume downloads to verified Recruiter accounts and shall log every download.	§5 5.2 Security
NFR-015	Shall	The system shall isolate AI service calls in a dedicated module such that AI provider failures do not raise unhandled errors in core request paths.	§5 5.3 Reliability
NFR-016	Shall	The system shall implement automatic retry with exponential backoff for transient failures in background job processing.	§5 5.3 Reliability
NFR-017	Shall	The system shall maintain a dead-letter queue for background jobs that exhaust retry attempts, surfaced to the Super Admin for manual review.	§5 5.3 Reliability
NFR-018	Shall	The system shall target 99.5% monthly uptime for core (non-AI) platform functionality.	§5 5.4 Availability
NFR-019	Shall	The system shall use cloud-based monitoring with automated alerting for service-down conditions.	§5 5.4 Availability
NFR-020	Shall	The system shall support horizontal auto-scaling of backend	§5 5.4 Availability

ID	Priority	Description	Location
		service instances in response to load.	
NFR-021	Shall	The system's database schema shall support tenant partitioning strategies (e.g., indexed college_id/company_id) to sustain growth without cross-tenant query degradation.	§5 5.5 Scalability
NFR-022	Shall	The system shall support caching of frequently read, infrequently changed data (e.g., department lists, platform taxonomies) in Redis with explicit invalidation on write.	§5 5.5 Scalability
NFR-023	Shall	The system's codebase shall be organized into clearly bounded modules per domain (e.g., Coordinators, Listings, Applications, AI Services) with business logic isolated to the service layer.	§5 5.6 Maintainability
NFR-024	Shall	The system shall maintain structured JSON logging with request/actor/tenant correlation identifiers on every mutating operation.	§5 5.6 Maintainability
NFR-025	Should	The system shall maintain automated test coverage for all service-layer business logic, per the strategy in Section 10.1.	§5 5.6 Maintainability
NFR-026	Shall	The system's web interface shall be responsive down to a 360px viewport width for all primary user-facing flows.	§5 5.7 Usability
NFR-027	Shall	The system shall present field-level validation errors inline on forms, mapped from backend validation responses.	§5 5.7 Usability
NFR-028	Should	The system shall provide empty-state guidance with a clear call-to-action on all primary list views.	§5 5.7 Usability
NFR-029	Shall	The system's backend shall be containerized (Docker) to allow deployment across compatible cloud hosts without code changes.	§5 5.8 Portability
NFR-030	Should	The system's frontend shall be deployable to any static/edge hosting provider supporting Next.js-equivalent React server rendering or a standard React SPA build.	§5 5.8 Portability
NFR-031	Shall	The system shall conform to WCAG 2.1 Level AA for primary user-facing flows, including keyboard navigability and screen-reader-compatible labeling.	§5 5.9 Accessibility
NFR-032	Shall	The system shall trap focus within modal dialogs and announce status changes via ARIA live regions.	§5 5.9 Accessibility